

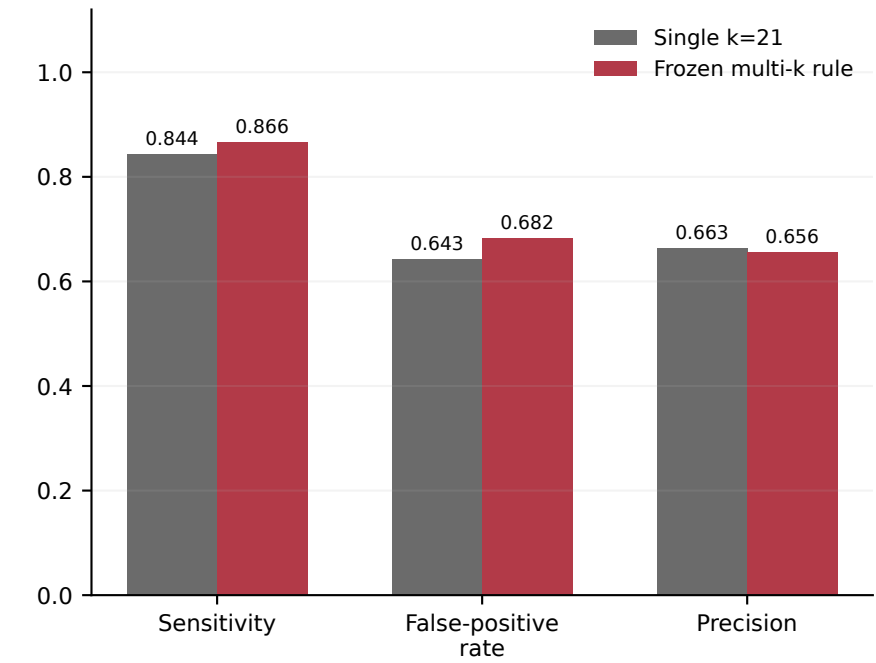
# Frozen collapse rule fails to control false positives under array divergence

## A Predeclared domain-shift design

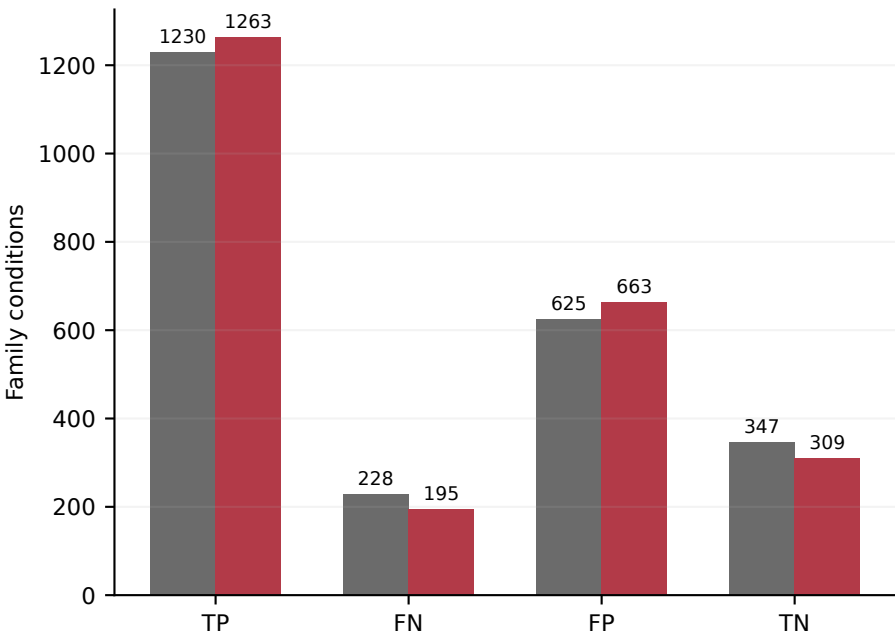
|                            |              |                              |
|----------------------------|--------------|------------------------------|
| <b>Fresh genomes</b>       | <b>3</b>     | seeds 5401-5403              |
| <b>Unit divergence</b>     | <b>3</b>     | 1%, 3%, 5% substitutions     |
| <b>Array structures</b>    | <b>2</b>     | one or three segments        |
| <b>Paired conditions</b>   | <b>2,430</b> | all baseline and frozen rows |
| <b>Multi-k unavailable</b> | <b>270</b>   | retained as NA               |

Frozen exact-copy rule applied without domain-shift fitting

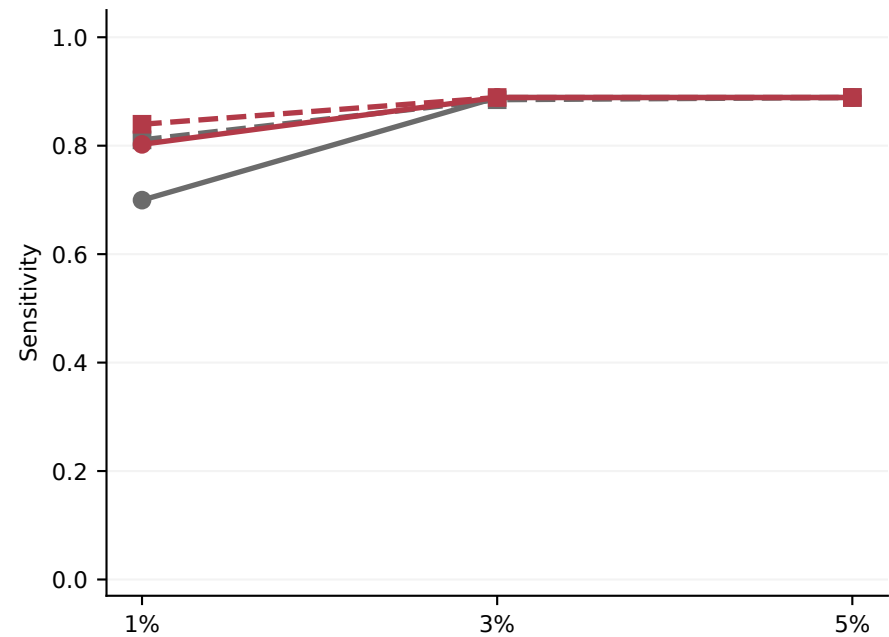
## B Overall domain-shift metrics



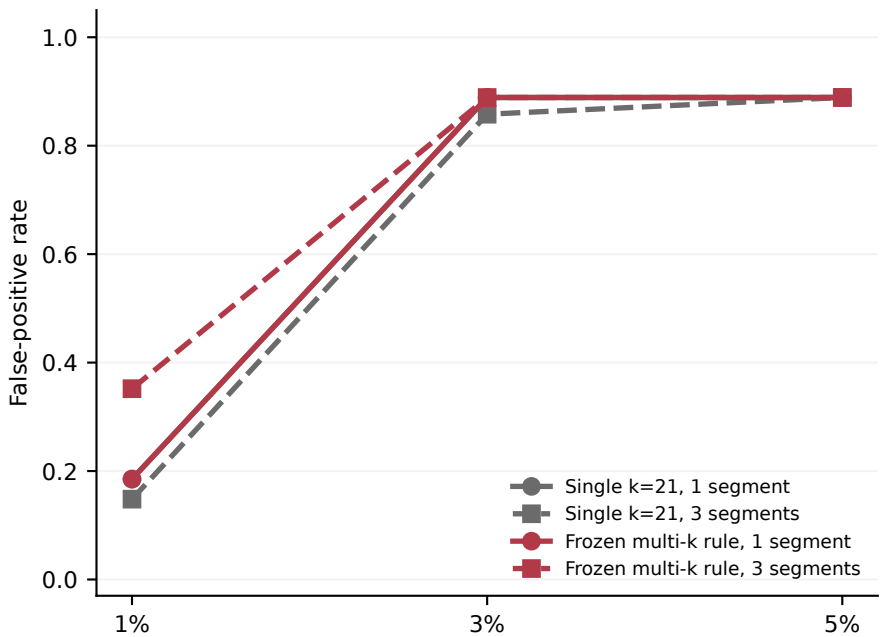
## C Confusion counts



## D Sensitivity by divergence



## E False-positive rate by divergence



## F Full-assembly localization

